A Flux-Driven Compact-to-Continuous Transition for a Magnetic Grushin Cylinder

Route-A source-local certificate HCS-C293

2 September 2026

Abstract

We give a complete channel-level spectral atlas for the magnetic Baouendi–Grushin form on $\mathbb{R} \times S^1$. Noninteger flux has compact resolvent. At integer flux exactly one angular channel becomes the free line Laplacian: its absolutely continuous spectrum coexists with embedded positive-integer oscillator eigenvalues, while the singular-continuous part is empty.

1 Friedrichs realization and conventions

Use $L^2(\mathbb{R} \times S^1, \mathrm{d}x\mathrm{d}\theta/(2\pi))$ and write $D_\theta = -i\partial_\theta$. For $\alpha \in \mathbb{R}$, begin with

$$q_\alpha[u] = \int_{\mathbb{R}\times S^1} \left(|\partial_x u|^2 + x^2|(D_\theta + \alpha)u|^2\right) \frac{\mathrm{d}x\mathrm{d}\theta}{2\pi}. \tag{1}$$

The form domain is the completion of compactly supported smooth functions in $\|u\|_2^2 + q_\alpha[u]$. Since (1) is densely defined, nonnegative, and closable, its closure determines a unique nonnegative self-adjoint operator by the representation theorem. We call it G_α. Formally,

$$\mathsf{G}_\alpha = -\partial_x^2 + x^2(D_\theta + \alpha)^2.$$

This is a Friedrichs-form statement. We do not assert essential self-adjointness of a smaller differential core.

Boscain, Prandi, and Seri study spectral and magnetic effects on related almost-Riemannian Grushin manifolds [1]; their geometric Laplace–Beltrami realization is not silently identified with the present Lebesgue-space form. Harakeh and Hillairet analyze Fourier-separated Baouendi–Grushin cylinder sectors and compact-resolvent mechanisms [2]. Our theorem is a self-contained source-model closure, not a claim that these classical mechanisms are newly discovered.

2 Fourier–Hermite reduction

The unitary angular Fourier expansion $u(x,\theta) = \sum_{k\in\mathbb{Z}} u_k(x)e^{ik\theta}$ gives

$$\mathsf{G}_\alpha \cong \bigoplus_{k\in\mathbb{Z}} h_{k+\alpha}, \qquad h_\omega = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \omega^2 x^2. \tag{2}$$

For $\omega \neq 0$, scaling the Hermite basis yields

$$\lambda_{k,n}(\alpha) = (2n+1)|k+\alpha|, \qquad k \in \mathbb{Z}, \quad n \in \mathbb{N}_0. \tag{3}$$

For $\omega = 0$, $h_0 = -\mathrm{d}^2/\mathrm{d}x^2$ is the free line Laplacian.

Theorem 1 (Complete flux dichotomy). *If $\alpha \notin \mathbb{Z}$, then G_α has compact resolvent and pure point spectrum given by (3), with multiplicities obtained by counting equal index pairs.*

If $\alpha \in \mathbb{Z}$, exactly one resonant angular Fourier channel is unitarily equivalent to the free line Laplacian. It contributes absolutely continuous spectrum $[0,\infty)$ of almost-everywhere multiplicity two. The orthogonal nonresonant sector has compact resolvent and point spectrum equal to every positive integer. Those eigenvalues are embedded in the full continuum. The singular-continuous spectrum is empty.

Proof. If $\delta = \operatorname{dist}(\alpha, \mathbb{Z}) > 0$, every channel is an oscillator and (3) is complete. For fixed L, the inequality $(2n+1)|k+\alpha| \leq L$ permits only finitely many k and, for each k, only finitely many n. Thus the eigenvalue list has finite multiplicities and tends to infinity, which is equivalent to compact resolvent.

At integer flux, a gauge shift reduces to $\alpha = 0$. The $k = 0$ summand in (2) is the free line Laplacian. Fourier transformation in x shows that its spectrum is purely absolutely continuous on $[0, \infty)$; the two momentum branches $\xi = \pm\sqrt{E}$ give multiplicity two for almost every $E > 0$. Every $k \neq 0$ summand is an oscillator, and the same finite-list criterion makes their direct sum compact-resolvent. A direct sum of this pure-point part and the free absolutely continuous part has no singular-continuous component. Formula (3) becomes $(2n+1)|k|$, so its values are positive integers and lie inside the free continuum. This proves coexistence, rather than incorrectly replacing the full integer-flux spectrum by either component alone. □

Flux shifts $\alpha \mapsto \alpha + j$, $j \in \mathbb{Z}$, merely relabel k. Complex conjugation and $k \mapsto -k$ give reflection $\alpha \mapsto -\alpha$. Hence the spectral fundamental interval is $0 \leq \delta \leq 1/2$.

3 Multiplicity and flux boundaries

At zero flux, a positive integer N occurs when $N = (2n+1)|k|$. Let $d_{\text{odd}}(N)$ count the positive odd divisors of N.

Proposition 1 (Exact embedded multiplicity)**.** *On the zero-flux nonresonant sector,*

$$\operatorname{mult}(N) = 2d_{\text{odd}}(N), \qquad N \geq 1. \tag{4}$$

Proof. Choosing an odd divisor $j = 2n+1$ fixes $|k| = N/j$. The two signs of k give two orthogonal angular modes, and every index pair arises uniquely in this way. □

If α is irrational, equality between two values $(2n+1)|k+\alpha|$ forces the two odd factors and signed channels to agree; otherwise it would make a nonzero integer multiple of α integral. Thus all off-integer-flux eigenvalues are simple for irrational flux. At rational flux the same equality is a finite Diophantine condition and may produce coincidences. Half flux has the systematic pairing $k \leftrightarrow -k-1$, so every level is at least double.

As α approaches an integer, the nearest-channel frequency $\delta = \operatorname{dist}(\alpha, \mathbb{Z})$ tends to zero. Its levels $(2n+1)\delta$ collapse, compactness is lost in the limit, and the channel becomes the free Laplacian. This singular transition is not uniform in heat trace or resolvent norm.

4 Heat traces

For $t > 0$ and $\omega > 0$, the oscillator trace is the geometric sum

$$\sum_{n \geq 0} e^{-t(2n+1)\omega} = \frac{1}{2\sinh(t\omega)}. \tag{5}$$

Positive-term summation is legitimate throughout.

Theorem 2 (Trace formulas)**.** *For $\alpha \notin \mathbb{Z}$,*

$$\operatorname{Tr} e^{-t\mathsf{G}_\alpha} = \sum_{k \in \mathbb{Z}} \frac{1}{2\sinh(t|k+\alpha|)} < \infty. \tag{6}$$

At integer flux the full heat operator is not trace class. On the orthogonal complement of the resonant channel,

$$\operatorname{Tr}_\perp e^{-t\mathsf{G}_0} = \sum_{m \geq 1} \frac{1}{\sinh(tm)} < \infty. \tag{7}$$

Moreover, as $\delta \downarrow 0$, the closest-channel contribution is $(2\sinh(t\delta))^{-1} \sim (2t\delta)^{-1}$.

Proof. Sum (5) over all Fourier channels. Away from integer flux, $|k+\alpha|$ grows linearly in $|k|$, so the outer sum converges. At zero flux, remove $k = 0$ and pair $k = \pm m$ to obtain (7). The final asymptotic follows from $\sinh z \sim z$. □

References

[1] U. Boscain, D. Prandi, and M. Seri, "Spectral analysis and the Aharonov–Bohm effect on certain almost-Riemannian manifolds," *Commun. Partial Differential Equations* 41 (2016), 32–50, arXiv:1406.6578.

[2] M. Harakeh and L. Hillairet, "A spectral condition for the control of eigenfunctions of Baouendi–Grushin type operators," arXiv:2312.04359 (2023).